

Freight rate prediction

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Data and cleaning

The development file contains 48,000 labeled loads from January-October 2025. The final set contains 12,000 unlabeled loads from November-December. IDs are unique, dates parse correctly, and validation IDs match the prediction template.

Training data has 300 missing weights, 292 non-positive weights and 374 missing market indices. Non-positive weights are treated as missing. Numeric medians and category encodings are fitted on training rows only. Original target values are retained; the maximum rate is \$25,533 versus a 99th percentile near \$5,973.

Chronological validation

Purpose	Period	Rows
Initial training	January-August 2025	38,477
Feature-set selection	September 2025	4,670
Holdout evaluation	October 2025	4,853

September MAE selects the feature set. The selected model is then refitted on January-September and tested on October. The external prediction set is not used for model selection. After evaluation, the final model is refitted on all 48,000 labeled loads.

Model and features

HistGradientBoostingRegressor is compared with an equipment-wise median rate-per-mile baseline. Three fixed feature sets are tested. The core set uses pickup, delivery, equipment, distance, weight, weekday and cyclic day-of-year sine/cosine features. Extended sets add coordinates and market index, then quote signal. The core set wins on September and is available for both final loads and the December inputs.

The model predicts rate per mile with absolute-error loss and distance sample weights, then multiplies predictions by distance to return dollars. This weighting makes the training objective equivalent to absolute dollar error. Categories are encoded for native categorical splits; missing numeric values include indicator flags. Parameters: 250 iterations, learning rate 0.08, 20 leaf nodes, minimum 40 samples per leaf, L2 regularization 2, seed 42; early stopping is disabled.

Recorded validation results

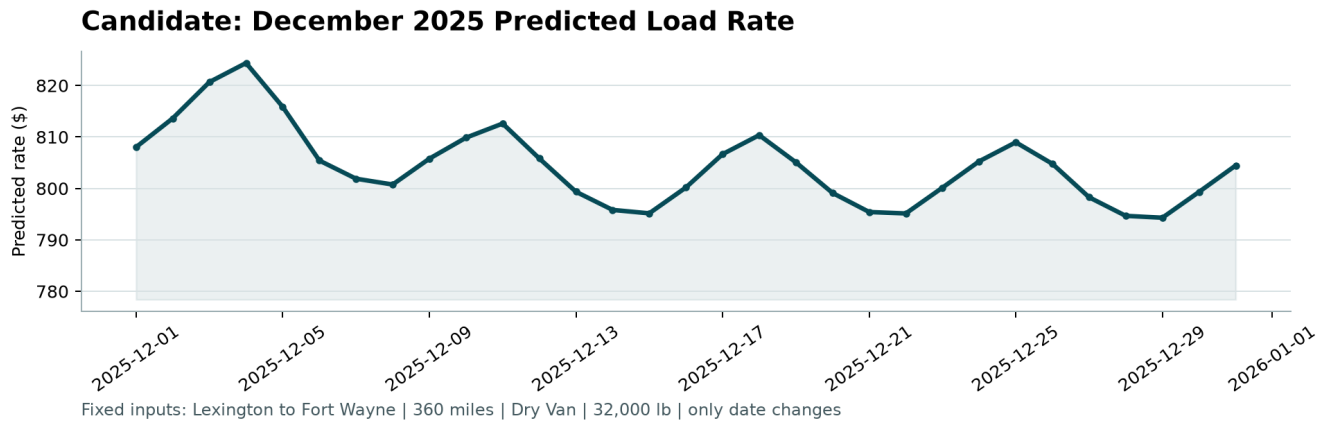
September model	MAE (\$)	RMSE (\$)	R2
Core	108.136	617.820	0.836
Extended + quote	114.640	619.104	0.835
Extended	121.288	621.333	0.834
Equipment median	227.323	657.066	0.814

October model	MAE (\$)	RMSE (\$)	R2
Selected core	117.935	650.518	0.819
Equipment median	231.305	683.865	0.800

December forecast and delivery

Fixed December prediction chart

The supplied, unchanged score.py generated the chart below from 31 predicted daily rates. Pickup is Lexington, delivery is Fort Wayne, distance is 360 miles, equipment is Dry Van and weight is 32,000 lb. Only the date changes.



The visible weekly pattern reflects the model's calendar features. This is a forecast for a fixed route, not a comparison with observed December prices. No missing geographic or market features are invented: the selected core model uses the fields available in both input formats.

Results and limits

October MAE improves by about 49% over the baseline. RMSE is much larger than MAE, indicating that substantial errors remain for some loads. R2 is 0.819; it is not a percentage accuracy score. Selection prioritizes MAE, while RMSE and R2 are also reported to show the tradeoff.

The model has only January-October labeled data, so seasonal behavior in November and December has not been directly validated. More chronological folds and analysis of large residuals would be useful follow-up work, but were not part of this submitted run.

Files and reproducibility

explore.py checks data quality and input IDs. **train.py** compares the fixed variants and saves development metrics. **predict.py** refits the selected model and writes validation_predictions.csv in template order, december_predictions.csv and the model artifact. **README.md** contains setup and run commands; **requirements.txt** pins the main package versions from the recorded run.

The recorded Windows run used Python 3.14.3 and scikit-learn 1.9.1. Metrics in this report are rounded to three decimals from that run. The original CSV inputs are preserved.

Output checks

The supplied scorer validated exactly 12,000 final predictions and 31 fixed December predictions and produced candidate_december.png. It checks structure, IDs, fixed inputs and positive finite rates; it does not calculate final accuracy. Spotter calculates final validation metrics after submission.

Sources: supplied assessment files, recorded candidate console results, and the December chart generated by the supplied scorer.